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**Long COVID is Associated with Lower Percentages of Mature, Cytotoxic NK Cell Phenotypes**

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**To the Editor:** Following SARS-CoV-2 infection, approximately 5% of individuals develop Long COVID (LC), defined as ongoing symptoms present at least 3 months later that are disruptive to everyday functioning. There is growing evidence that SARS-CoV-2 persistence is associated with LC, and that people with LC have dysregulated adaptive immune function that may originate from or potentiate viral persistence (1). Natural Killer (NK) cells, a critical component of the viral innate immune response, can exhibit dysfunctional phenotypes during acute COVID-19, and these alterations may hinder clearance of SARS-CoV-2 infected cells. COVID-19 can also induce adaptive NK cell responses during acute infection (2, 3) but there is a critical lack of knowledge about NK cell phenotypes in the post-acute phase and their relationship to LC.

We performed high-dimensional NK cell phenotyping on PBMCs collected 4 months following SARS-CoV-2 infection from people with LC (N=21) and those who had prior COVID-19 but fully recovered (N=14) in the UCSF LIINC cohort (4). Samples were collected in the first year of the pandemic prior to the emergence of vaccination, antiviral treatment, or frequent reinfection, and most participants (74%) had mild/moderate outpatient COVID-19 (Table S1). LC was defined as the presence of COVID-attributed symptoms at least 3 months following the initial infection, severe LC was defined as the presence of >5 symptoms, and all participants were confirmed to have consistent presence or absence of LC symptoms for up to 8 months. We examined associations between NK cell phenotypes, demographic and LC clinical factors, and SARS-CoV-2 antigen persistence.

Age was similar across groups (Fig 1A), but a larger proportion of people with LC were female (Fig 1B), and those with LC were more symptomatic during acute infection (Fig 1C). Whereas percentages of total $CD56^{+}$ NK cells were similar across groups (Fig 1E), mature, cytotoxic $CD56^{dim}/CD16^{+}$ NK cell percentages were significantly lower in participants with LC compared to those who had fully recovered (P=0.037, Fig 1F); a more pronounced significant reduction was observed in those with severe LC (P=0.036; Fig 1G). $CD56^{dim}/CD16^{+}$ NK cells play an important role in clearing virus-infected cells from the body, and we observed a significant negative correlation between the percentage of these cells and the number of reported LC symptoms (Spearman r=-0.5, P=0.022, Fig 1H). We also observed significantly lower $CD56^{dim}/CD16^{+}$ NK cell percentages in participants reporting any neurocognitive and gastrointestinal symptoms or fatigue compared to recovered

controls (P=0.037, P=0.018, P=0.041; Fig 1I). Participants with severe LC also had significantly lower percentages of CD56$^{dim}$ NK cells expressing KIR2D(L1/S1/S3/S5) than the recovered group (P=0.036; Fig 1G) but we did not identify a significant relationship between NK cell phenotypes and post-acute plasma SARS-CoV-2 antigen detection (Fig S1).

Given the percentages of CD56$^{dim}$/CD16$^{+}$ NK cells were significantly lower in females than males (Fig 1J), we repeated comparisons of NK cell phenotypes including only females and observed a significantly lower percentage of CD56$^{dim}$/CD16$^{+}$ (47.1% vs 63.5%) and CD56$^{dim}$/KIR2D(L1/S1/S3/S5)$^{+}$ (16.2% vs 29.2%) NK cell populations in those with severe LC versus those who fully recovered (P=0.045 and 0.011, respectively; Figs 1K, S1). These data suggest that the differences observed implicate underlying factors between NK cell phenotypes and LC in addition to potential sex differences.

Natural killer cells have been shown to acquire adaptive-like subsets and can be distinguished by their expression of NKG2C and loss of FcεR1γ and PLZF. Therefore, we performed a multidimensional analysis (Fig 1L) to define the relationship between adaptive NK cell phenotypes and LC, as well as with other factors that may modulate LC, such as CMV seropositivity and the presence of EBV early antigen D IgG (a surrogate for recent EBV reactivation). Although we did not observe any significant differences in NK cell UMAP clusters with adaptive phenotypes between those with and without any LC phenotype, SARS-CoV-2 antigen detection, or EBV antibodies, we did observe a significant increase in the percentage of adaptive NK clusters in CMV IgG$^{+}$ participants (n=18) versus IgG$^{-}$ participants (n=17) (P=0.024; Figs 1M, S1).

In conclusion, we observed that the proportion of NK cells with a mature and cytotoxic phenotype was significantly lower in people with LC compared to those who fully recovered. These cells express CD16, an IgG receptor that mediates antibody-dependent cytotoxicity, and although we did not observe a direct association between post-acute plasma SARS-CoV-2 antigen detection and these NK cell phenotypes, it is possible that reduced cytotoxicity during initial tissue infection or later clearance of persistently infected cells may be related to the development of LC. Alternatively, these data could suggest tissue redistribution of mature, cytotoxic NK

cell subsets to tissues where antigen persists. Further study of the relationships between cytotoxic immune responses, viral persistence, and LC is urgently needed.

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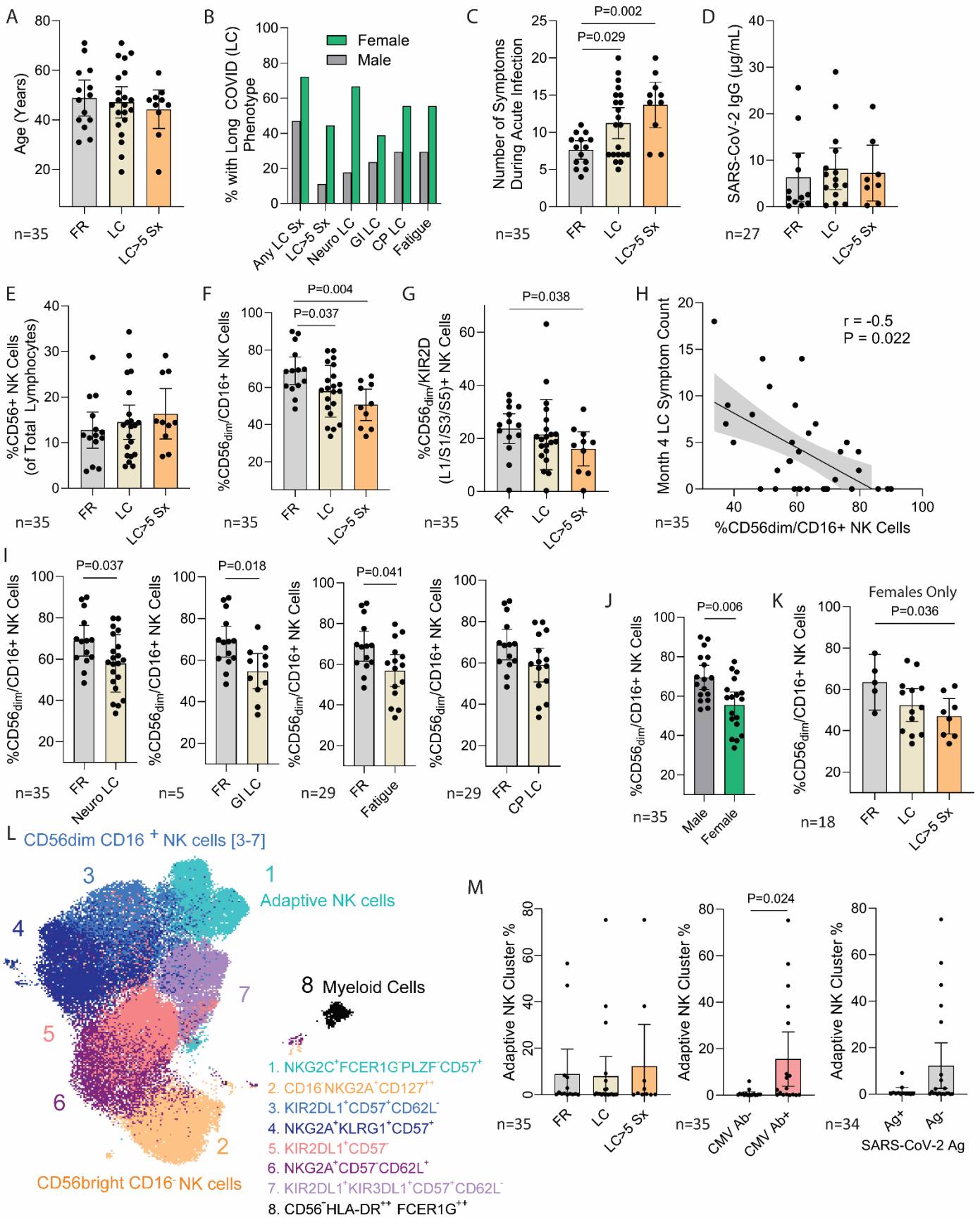
A
Age (Years)
n=35
FR
LC
LC>5 Sx
B
% with Long COVID (LC) Phenotype
Female
Male
Any LC Sx
LC>5 Sx
Neuro LC
GI LC
CP LC
Fatigue
C
Number of Symptoms During Acute Infection
P=0.002
P=0.029
n=35
D
SARS-CoV-2 IgG (µg/mL)
n=27
E
%CD56+ NK Cells (of Total Lymphocytes)
n=35
F
%CD56dim/CD16+ NK Cells
P=0.004
P=0.037
n=35
G
%CD56dim/KIR2D (L1/S1/S3/S5)+ NK Cells
P=0.038
n=35
H
Month 4 LC Symptom Count
r = -0.5
P = 0.022
%CD56dim/CD16+ NK Cells
n=35
I
%CD56dim/CD16+ NK Cells
P=0.037
n=35
Neuro LC
P=0.018
n=5
GI LC
P=0.041
n=29
Fatigue
n=29
CP LC
J
P=0.006
Male
Female
n=35
K
Females Only
P=0.036
n=18
L
CD56dim CD16 + NK cells [3-7]
1 Adaptive NK cells
2
3
4
5
6
7
8 Myeloid Cells
CD56bright CD16- NK cells
1. NKG2C+FCER1G-PLZF-CD57+
2. CD16-NKG2A+CD127++
3. KIR2DL1+CD57+CD62L-
4. NKG2A+KLRG1+CD57+
5. KIR2DL1+CD57-
6. NKG2A+CD57-CD62L+
7. KIR2DL1+KIR3DL1+CD57+CD62L-
8. CD56-HLA-DR++ FCER1G++
M
Adaptive NK Cluster %
n=35
P=0.024
CMV Ab-
CMV Ab+
n=34
Ag+
Ag-
SARS-CoV-2 Ag

**Fig 1. Relationships between NK cell phenotypes and Long COVID**. Age (A), the percentage of male and female participants (B), the number of symptoms (sx) during acute infection (C), mean SARS-CoV-2 total anti-receptor binding domain (RBD) IgG levels (D) along with percentages of total $CD56^{+}$ NK cells (E), $CD56^{dim}/CD16^{+}$ NK cells (F), and $CD56^{dim}$ NK cells expressing KIR2D(L1/S1/S3/S5) (G) approximately four months following acute infection in those with Long COVID (LC) and those who fully recovered. Spearman correlation analysis of $CD56^{dim}/CD16^{+}$ NK cell percentages and the number of LC symptoms (H), and the percentage of $CD56^{dim}/CD16^{+}$ NK cells in participants with neurocognitive (neuro), gastrointestinal (GI), fatigue, and cardiopulmonary (CP) LC (I) are shown. $CD56^{dim}/CD16^{+}$ NK cells percentages in male and female participants (J) and repeat $CD56^{dim}/CD16^{+}$ comparisons including only female participants (K). NK cell cluster UMAP (L) and the percent of adaptive NK cell clusters by LC symptoms group, CMV IgG serostatus, and presence or absence of SARS-CoV-2 spike or nucleocapsid antigen in blood after COVID-19 are shown (M). Mean (bars) and 95% confidence intervals are shown. P-values from 2-tailed Mann-Whitney U testing (I, J, M) or Kruskal-Wallis testing (C, F, G, K).